CHAPTER 10

BAIRE SPACE

Next to the natural numbers, perhaps the most fundamental object of study of set theory is **Baire space**,

$$\mathcal{N} =_{\text{df}} (\mathbb{N} \to \mathbb{N}), \tag{10-1}$$

the set of all number theoretic sequences. If we let

$$\mathcal{C} =_{\text{df}} (\mathbb{N} \to \{0, 1\}) \tag{10-2}$$

be the **Cantor set**[23] of all infinite, binary sequences, then

$$\mathcal{C} \subseteq \mathcal{N} \subseteq \mathcal{P}(\mathbb{N} \times \mathbb{N}),$$

and with now familiar computations,

$$\mathfrak{c} =_c 2^{\aleph_0} =_c |\mathcal{P}(\mathbb{N})| =_c |\mathcal{C}| \leq_c |\mathcal{N}| \leq_c |\mathcal{P}(\mathbb{N} \times \mathbb{N})| =_c |\mathcal{P}(\mathbb{N})| = \mathfrak{c}.$$

Since $\mathcal{N} =_c \mathbb{R}$ will follow as in Chapter **2** from the proper definitions in Appendix **A**, the Continuum Hypothesis **3.2** is equivalent to the proposition

$$(\textbf{CH}) \qquad (\forall X \subseteq \mathcal{N})[X \leq_c \mathbb{N} \vee X =_c \mathcal{N}].$$

In fact, there is such a tight connection between $\mathcal{N}$, $\mathcal{C}$ and $\mathbb{R}$ that practically every interesting property of one of these spaces translates immediately to a related, interesting property of the others. In the problems we will make this precise for $\mathcal{N}$ and $\mathcal{C}$ and in Appendix **A** for $\mathbb{R}$, where we will also draw the consequences of the results of this chapter for the real numbers.[24]

The material in this chapter is not necessary for the comprehension of the two chapters which follow, and the exposition is more condensed and requires more effort from the reader than the rest of these Notes. In a first reading, it may be best to skip it and come back to it after Chapters 11 and 12 have been mastered.

[23] It is traditional to use the same name for this subset of $\mathcal{N}$ and the set of real numbers defined in the proof of **2.14**. Figure **2.4** explains vividly the reason for this, and nobody has ever been confused by it.

[24] One may think of $\mathcal{N}$ as a "discrete", "digital", or "combinatorial" version of the "continuous" or "analog" $\mathbb{R}$. A real number x is completely determined by a decimal expansion $x(0) \,.\, x(1)x(2)\ldots$, where $(n \mapsto x(n)) \in \mathcal{N}$, *but* two distinct decimal expansions may compute to the same real number. This is a big "but", it is the key fact behind the so-called *topological connectedness* of the real line which is of interest in analysis, to be sure, but of little set theoretic consequence. We may view Baire space as a "digital version" of $\mathbb{R}$ because it does not make any such identifications, each point $x \in \mathcal{N}$ determines unambiguously its "digits" $x(0), x(1), \ldots$.

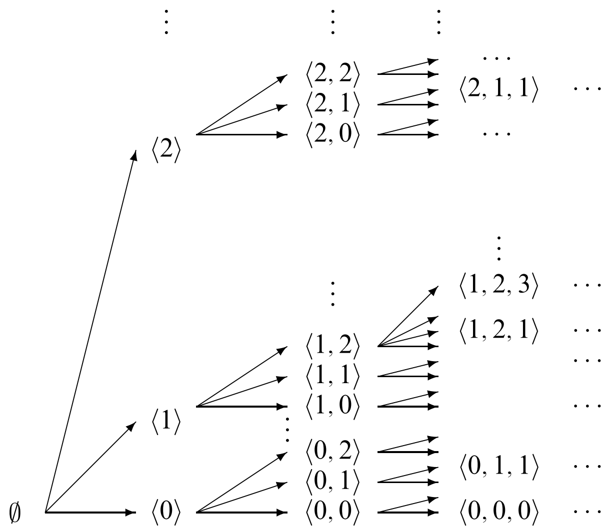

FIGURE **10.1**. A small part of Baire space.

Our aim here is to establish some elementary facts about $\mathcal{N}$ which bear on the Continuum Problem. We will define the family of ANALYTIC SUBSETS OF $\mathcal{N}$ and prove that every analytic set satisfies the Continuum Hypothesis, in the sense that it must be either countable or equinumerous with $\mathcal{N}$. This PERFECT SET THEOREM **10.20** is significant because essentially every set of interest in classical analysis is analytic, including all the BOREL SETS which play a fundamental role in measure theory and integration; by SUSLIN'S THEOREM **10.31**, a set $A \subseteq \mathcal{N}$ is Borel exactly when both A and $\mathcal{N} \setminus A$ are analytic. On the other hand, we will show in Theorem **10.32** that the basic and natural method of proving the Continuum Hypothesis for analytic sets cannot be extended to solve the full Continuum Problem, which remains open. In addition to their applications in analysis, Theorems **10.20** and **10.32** are of substantial foundational interest, as their proofs illustrate beautifully the role of choice principles in classical mathematics.

10.1. The structure of $\mathcal{N}$. Our intuitions about $\mathcal{N}$ come from picturing it as the body of the largest tree on $\mathbb{N}$ in the terminology of **9.3** and (9-3),

$$\mathcal{N} = [\mathbb{N}^*].$$

We will refer to subsets of Baire space as **pointsets**, the term "point" temporarily reserved for members of $\mathcal{N}$, infinite branches in $\mathbb{N}^*$. By the **complement** of a pointset, we will mean its complement in $\mathcal{N}$,

$$cA =_{\mathrm{df}} \mathcal{N} \setminus A. \qquad (10\text{-}3)$$

It is convenient to extend the initial segment notation on strings,

$$u \sqsubseteq x \iff_{\mathrm{df}} u \subset x \quad (u \in \mathbb{N}^*, x \in \mathcal{N}), \qquad (10\text{-}4)$$

to indicate that a finite sequence u is an initial segment of the point x, an **approximation** of x which determines the first $\mathrm{lh}(u)$ values of x. For each $u \in \mathbb{N}^*$, the set

$$\mathcal{N}_u =_{\mathrm{df}} \{x \in \mathcal{N} \mid u \sqsubseteq x\} = [\mathbb{N}^*_u] \tag{10-5}$$

of points in $\mathcal{N}$ which extend u is the **neighborhood** determined by u in $\mathcal{N}$.

10.2. Exercise. *For all $u, v \in \mathbb{N}^*$,*

$$u \sqsubseteq v \iff \mathcal{N}_v \subseteq \mathcal{N}_u.$$

10.3. Exercise. *The family of neighborhoods is countable.*

10.4. Definition. A pointset G is **open** if it is a union of neighborhoods, so that

$$x \in G \iff (\exists u)[x \in \mathcal{N}_u \;\&\; \mathcal{N}_u \subseteq G]; \tag{10-6}$$

closed if its complement is open; and **clopen** if it is both closed and open.

Open sets are often defined by the following, easily equivalent condition:

10.5. Exercise. *A set $G \subseteq \mathcal{N}$ is open if and only if for every $x \in G$, there exists some neighborhood $\mathcal{N}_u$ such that $x \in \mathcal{N}_u \subseteq G$.*

10.6. Proposition. (1) *$\emptyset$, $\mathcal{N}$ and all neighborhoods are clopen.*

(2) *Every singleton $\{x\}$ is closed but not open.*

(3) *Every non-empty open pointset is the union of a sequence of neighborhoods.*

(4) *The union $\bigcup \mathcal{G}$ of a family $\mathcal{G}$ of open pointsets is open and, dually, the intersection $\bigcap \mathcal{F}$ of a family $\mathcal{F}$ of closed pointsets is closed.* (We assume $\bigcap \emptyset = \mathcal{N}$, so the intersection operation is defined for every family of pointsets.)

(5) *The intersection $G_1 \cap G_2$ of two open pointsets is open, and dually, the union $F_1 \cup F_2$ of two closed pointsets is closed.*

PROOF. (1) Each neighborhood is open, since $\mathcal{N}_u = \bigcup\{\mathcal{N}_u\}$, and, in particular, $\mathcal{N} = \mathcal{N}_{\langle\,\rangle}$ is open. Neighborhoods are also closed by Exercise **10.5**: if $x \notin \mathcal{N}_u$, then $u \not\sqsubseteq x$, so there exists some $i < \mathrm{lh}(u)$ such that $x(i) \neq u(i)$ and then $x \in \mathcal{N}_{\langle x(0),\dots,x(i)\rangle}$ while $\mathcal{N}_{\langle x(0),\dots,x(i)\rangle} \cap \mathcal{N}_u = \emptyset$. The empty set is the union of the empty (!) family of neighborhoods, formally

$$\emptyset = \bigcup\{\mathcal{N}_u \mid \mathcal{N}_u \subseteq \emptyset\}.$$

(2) A singleton $\{x\}$ is not open, because it does not contain any neighborhood, and so it cannot be a union of neighborhoods. Its complement is open, however, since

$$\mathcal{N} \setminus \{x\} = \bigcup\{\mathcal{N}_u \mid u \not\sqsubseteq x\}.$$

(3) If G is open and non-empty, then the set $\{u \mid \mathcal{N}_u \subseteq G\}$ is non-empty and countable, and so it can be enumerated.

(4) This is immediate from the characterization of open sets in Exercise **10.5**.

(5) If G_1, G_2 are open and $x \in G_1 \cap G_2$, then there exist $u, v \sqsubseteq x$ such that $\mathcal{N}_u \subseteq G_1$ and $\mathcal{N}_v \subseteq G_2$. The finite sequences u, v are comparable since they are both initial segments of x, so suppose $u \sqsubseteq v$, the argument being the same

in the opposite case: now $\mathcal{N}_u \supseteq \mathcal{N}_v$, so $\mathcal{N}_v \subseteq G_1 \cap G_2$, as required. The dual property for closed sets follows by taking complements. $\dashv$

Baire space is a *topological space* by the classical definition recounted in **4.30**, but a very special one, because of the next, basic connection between its topology and the combinatorial structure of the tree $\mathbb{N}^*$. In proving it—and in the sequel, routinely—we will use the following trivial equivalence relating a tree T and its body:

$$x \in [T] \iff (\forall u \sqsubseteq x)[u \in T]. \tag{10-7}$$

It follows immediately from the definition of $[T]$, (9-3).

10.7. Proposition. *A pointset F is closed if and only if it is the body of a tree T on $\mathbb{N}$, $F = [T]$.*

PROOF. If $x \notin [T]$, then for some $u \sqsubseteq x$, $u \notin T$, and then $\mathcal{N}_u \cap [T] = \emptyset$, so $\mathcal{N}_u \subseteq c[T]$; thus $c[T]$ is open and $[T]$ is closed. Conversely, if we associate with each pointset F the tree

$$T^F =_{\mathrm{df}} \{u \in \mathbb{N}^* \mid (\exists x \in F)[u \sqsubseteq x]\}, \tag{10-8}$$

then obviously

$$F \subseteq [T^F].$$

If F is closed, we also have $[T^F] \subseteq F$: because if $x \notin F$, then for some $u \sqsubseteq x$, $\mathcal{N}_u \cap F = \emptyset$ by the openness of the complement cF, hence $u \notin T^F$ and $x \notin [T^F]$ by (10-7). $\dashv$

This basic characterization allows us to classify closed pointsets by the combinatorial properties of the trees which define them. It is not wrong to think of the cluster of combinatorial notions to come as the **combinatorial geometry** of $\mathcal{N}$, although it is not a "geometry" by any standard, classical definition of this term.

10.8. Definition. Set

$$\begin{aligned} u \mid v &\iff_{\mathrm{df}} u, v \text{ are incompatible} \\ &\iff (\exists i < \mathrm{lh}(u), \mathrm{lh}(v))[u(i) \neq v(i)], \end{aligned} \tag{10-9}$$

and, by extension,

$$u \mid x \iff_{\mathrm{df}} \neg[u \sqsubseteq x] \iff (\exists v \sqsubseteq x)[u \mid v].$$

A string u **splits** in a tree T if it has incompatible extensions in T and a tree T is **splitting** if every $u \in T$ splits in T,

$$u \in T \Longrightarrow (\exists u_1, u_2 \in T)[u \sqsubseteq u_1 \,\&\, u \sqsubseteq u_2 \,\&\, u_1 \mid u_2].$$

Notice that a splitting tree has no terminal nodes.

A pointset P is **perfect** *if it is the body of a splitting tree.* Perfect sets are automatically closed.

10.9. Exercise. *Each neighborhood $\mathcal{N}_u$ is perfect.*

10.10. Proposition. *Every non-empty, perfect pointset P has cardinality $\mathfrak{c}$.*

PROOF. Suppose $P = [T]$ with T non-empty, splitting, and choose functions

$$l : T \to T, \quad r : T \to T$$

which witness the splitting property for T, i.e., for each $u \in T$,

$$u \sqsubseteq l(u), \quad u \sqsubseteq r(u), \quad l(u) \mid r(u).$$

By the String Recursion Theorem **5.33**, there is a function

$$\sigma : \{0,1\}^* \to T$$

from the tree of all binary strings into T which satisfies the identities

$$\sigma(\emptyset) = \emptyset, \quad \sigma(u \star \langle 0 \rangle) = l(\sigma(u)), \quad \sigma(u \star \langle 1 \rangle) = r(\sigma(u)).$$

Thus $\sigma(u \star \langle i \rangle)$ is a proper extension of $\sigma(u)$ for $i = 0, 1$, so σ is strictly monotone,

$$u \sqsubsetneq v \Longrightarrow \sigma(u) \sqsubsetneq \sigma(v),$$

and we can define a function $\pi : \mathcal{C} \to [T]$ by

$$\pi(x) =_{\mathrm{df}} \sup \{\sigma(u) \mid u \sqsubseteq x\}. \tag{10-10}$$

The key property of σ is that it also preserves incompatibility,

$$u \mid v \Longrightarrow \sigma(u) \mid \sigma(v). \tag{10-11}$$

To see this, let i be least such that $u(i) \neq v(i)$, so for some w we have

$$w \star \langle 0 \rangle \sqsubseteq u, \quad w \star \langle 1 \rangle \sqsubseteq v$$

(or the other way around); now $\sigma(w \star \langle 0 \rangle)$ and $\sigma(w \star \langle 1 \rangle)$ are incompatible by their definition, and the monotonicity property implies that $\sigma(u)$, $\sigma(v)$ extend them, so they are incompatible too. Finally, (10-11) implies that π is an injection, and this establishes that $\mathcal{C} \leq_c [T]$, which is all we need. ⊣

This simple abstraction of Cantor's proof of the uncountability of the reals (**2.14**) suggests an attack on the Continuum Problem: to prove that an uncountable pointset has cardinality $\mathfrak{c}$, it is enough to show that it contains a non-empty, perfect subset. This is trivially true of open sets (because each $\mathcal{N}_u$ is perfect) and it is also true of closed sets, less trivially.

10.11. Cantor-Bendixson Theorem. *Every closed subset F of $\mathcal{N}$ can be decomposed uniquely into two disjoint subsets*

$$F = P \cup S, \quad P \cap S = \emptyset, \tag{10-12}$$

where P, the **kernel** *of F, is perfect and S, the* **scattered part** *of F, is countable.*

It follows that every uncountable, closed pointset has a non-empty, perfect kernel and hence has cardinality $\mathfrak{c}$.

PROOF. Let $T = T^F$ as in (10-8), so that T has no terminal nodes and $F = [T]$, and set

$$S =_{\mathrm{df}} \bigcup \{[T_u] \mid u \in T \;\&\; |[T_u]| \leq_c \aleph_0\},$$
$$P =_{\mathrm{df}} F \setminus S.$$

By its definition, S is the union of countably many, countable sets, so it is countable (note the use of $\mathbf{AC}_{\mathbb{N}}$ here), and it remains to show that P is perfect.

The set of strings

$$kT = \{u \in T \mid |[T_u]| >_c \aleph_0\}$$

is easily a tree, and

$$x \in S \iff x \in F \;\&\; (\exists u \sqsubseteq x)[u \notin kT]$$

is another way to read the definition of S. Since $P = F \setminus S$,

$$\begin{aligned} x \in P &\iff x \in F \;\&\; [x \notin F \vee (\forall u \sqsubseteq x)[u \in kT]] \\ &\iff (\forall u \sqsubseteq x)[u \in T] \;\&\; (\forall u \sqsubseteq x)[u \in kT] \\ &\iff (\forall u \sqsubseteq x)[u \in kT] \\ &\iff x \in [kT], \end{aligned}$$

and it is enough to prove that kT is splitting. Suppose, towards a contradiction that some $u \in kT$ does not split. This means that all extensions of u in kT are compatible and they define a single point

$$x = \sup\{v \in kT \mid u \sqsubseteq v\}.$$

Since every extension of u in kT approximates x,

$$[T_u] = \{x\} \cup \bigcup\{[T_v] \mid u \sqsubseteq v \in T \;\&\; |[T_v]| \leq_c \aleph_0\};$$

this, however, implies that $[T_u]$ is a countable union of countable sets, which is absurd.

We leave the uniqueness of the decomposition (10-12) for the problems, **x10.2**. ⊣

10.12. Definition. A family Γ of pointsets has **property P** if every uncountable set in Γ contains a non-empty, perfect subset. In this classical terminology,[25] the family $\mathcal{F}$ of closed pointsets has property **P**, or (more simply) *every closed pointset has property* **P**.

10.13. Exercise. *If a family of pointsets* Γ *has property* **P**, *then the family* Γ_σ *of all countable unions of sets in* Γ *also has property* **P**.

Thus every $\mathcal{F}_\sigma$ pointset, of the form

$$A = \bigcup\nolimits_{n \in \mathbb{N}} F_n \tag{10-13}$$

[25] The classical terminology in question is quite absurd, but so well established that it would be folly to change it or bypass it. In any topological space, closed sets are $\mathcal{F}$-sets, from the French *fermèt*; open sets are $\mathcal{G}$-sets, from the German *Gebiete* (it means *region*); countable unions of Γ-sets are Γ_σ-sets, and countable intersections of Γ-sets are Γ_δ-sets, from the German words *Summe* and *Durchschnit* for *union* and *intersection*, respectively. We will only use this terminology in passing references to $\mathcal{F}_\sigma$ and $\mathcal{G}_\delta$ pointsets.

with each F_n closed has property **P**. The same is true of every $\mathcal{G}_\delta$ set, of the form

$$A = \bigcap_{n \in \mathbb{N}} G_n \tag{10-14}$$

with each G_n open, but the proof is not that simple, and it is ultimately easier to establish directly the property **P** for the much larger family of analytic pointsets.

10.14. Definition. Recall from **6.25** that a function $f : X \to Y$ on one topological space to another is **continuous** if the inverse image $f^{-1}[G]$ of every open set in Y is open in X. *A pointset $A \subseteq \mathcal{N}$ is* **analytic** *or* **Suslin**, *if either $A = \emptyset$ or A is the image of Baire space under a continuous function*, in symbols,

$$\mathcal{A} =_{\text{df}} \{A \subseteq \mathcal{N} \mid A = \emptyset \vee (\exists \text{ continuous } f : \mathcal{N} \to \mathcal{N})[A = f[\mathcal{N}]]\}.$$

Continuity in $\mathcal{N}$ has a simple, combinatorial interpretation which is the key to its applications.

10.15. Theorem. *A function $f : \mathcal{N} \to \mathcal{N}$ is continuous if and only if there exists a monotone function $\tau : \mathbb{N}^* \to \mathbb{N}^*$ on strings, such that*

$$\begin{aligned} f(x) &= \sup\{\tau(u) \mid u \sqsubseteq x\} \quad (x \in \mathcal{N}) \\ &= \lim_n \tau(\overline{x}(n)). \end{aligned} \tag{10-15}$$

When $\tau : \mathbb{N}^* \to \mathbb{N}^*$ is monotone and (10-15) holds, we say that τ **computes** the function f.

PROOF. If f satisfies (10-15), then

$$f(x) \in \mathcal{N}_v \iff (\exists u \sqsubseteq x)[v \sqsubseteq \tau(u)],$$

so each inverse image of a neighborhood

$$f^{-1}[\mathcal{N}_v] = \bigcup\{\mathcal{N}_u \mid v \sqsubseteq \tau(u)\}$$

is a union of neighborhoods and f is continuous.

For the more difficult converse, suppose f is continuous and let

$$S(u) =_{\text{df}} \{v \in \mathbb{N}^* \mid f[\mathcal{N}_u] \subseteq \mathcal{N}_v\} \quad (u \in \mathbb{N}^*).$$

Each $S(u) \neq \emptyset$, since the root $\emptyset \in S(u)$; $v \sqsubseteq v' \in S(u) \Longrightarrow v \in S(u)$; and

$$v, v' \in S(u) \Longrightarrow f[\mathcal{N}_u] \subseteq \mathcal{N}_v \cap \mathcal{N}_{v'} \Longrightarrow [v \sqsubseteq v' \vee v' \sqsubseteq v],$$

since $v \mid v' \Longrightarrow \mathcal{N}_v \cap \mathcal{N}_{v'} = \emptyset$. Thus, there are two possibilities:

CASE 1. There is some $v \in S(u)$ such that $\text{lh}(v) = \text{lh}(u)$. In this case we set

$$\tau(u) =_{\text{df}} v = \text{ the unique string in } S(u) \text{ such that } \text{lh}(v) = \text{lh}(u).$$

CASE 2. There is no $v \in S(u)$ such that $\text{lh}(v) = \text{lh}(u)$. Now we set

$$\tau(u) =_{\text{df}} \sup\{v \mid v \in S(u)\}.$$

The monotonicity of τ follows easily from the implications

$$u_1 \sqsubseteq u_2 \Longrightarrow f[\mathcal{N}_{u_1}] \supseteq f[\mathcal{N}_{u_2}] \Longrightarrow S(u_1) \subseteq S(u_2),$$

considering the possibilities in the definitions of $\tau(u_1)$ and $\tau(u_2)$. To prove (10-15), notice first that because $\tau(u) \in S(u)$,

$$u \sqsubseteq x \in \mathcal{N} \Longrightarrow f(x) \in \mathcal{N}_{\tau(u)} \Longrightarrow \tau(u) \sqsubseteq f(x).$$

Moreover, by the continuity of f, if $v \sqsubseteq f(x)$, then there is some $u \sqsubseteq x$ such that $f[\mathcal{N}_u] \subseteq \mathcal{N}_v$, hence $v \in S(u)$ and either immediately $v \sqsubseteq \tau(u)$, if $\tau(u)$ is defined by CASE 2, or there is some u' extending u, with $\mathrm{lh}(u') = \mathrm{lh}(v)$ such that $v = \tau(u')$ in the other case. $\dashv$

It is useful to think of (10-15) as a computational characterization of continuity: the string function $\tau(u)$ gives us better and better approximations $\tau(u) \sqsubseteq f(x)$ to the value of f, as we feed into it successively finer approximations $u \sqsubseteq x$ to the argument. We can turn this picture into a precise and elegant result, in terms of the notions introduced in Chapter **6**.

10.16. Corollary. *A function $f : \mathcal{N} \to \mathcal{N}$ is continuous if and only if it is the restriction to $\mathcal{N}$ of some monotone, continuous mapping*

$$\pi : (\mathbb{N} \rightharpoonup \mathbb{N}) \to (\mathbb{N} \rightharpoonup \mathbb{N})$$

on the inductive poset $(\mathbb{N} \rightharpoonup \mathbb{N})$. By Definition **6.22**, a monotone mapping $\pi : (\mathbb{N} \rightharpoonup \mathbb{N}) \to (\mathbb{N} \to \mathbb{N})$ is continuous if it satisfies the equivalence

$$\pi(x)(i) = w \iff (\exists u \in \mathbb{N}^*)[u \sqsubseteq x \;\&\; \pi(u)(i) = w].$$

Here we are using the fact that $\mathcal{N}$ is a subset of $(\mathbb{N} \rightharpoonup \mathbb{N})$, consisting precisely of all its maximal elements, and the basic observation is the decomposition

$$(\mathbb{N} \rightharpoonup \mathbb{N}) = \mathbb{N}^* \cup \mathcal{N}, \quad \mathbb{N}^* \cap \mathcal{N} = \emptyset. \tag{10-16}$$

PROOF. If $f : \mathcal{N} \to \mathcal{N}$ is continuous, let τ compute it by the Theorem and take (literally)

$$\pi = \tau \cup f,$$

i.e., $\pi(u) = \tau(u)$ for $u \in \mathbb{N}^*$ and $\pi(x) = f(x)$ for $x \in \mathcal{N}$. The continuity of π is trivial. The converse is very easy. $\dashv$

10.17. Exercise. *Prove the "easy converse", i.e., that if $f : \mathcal{N} \to \mathcal{N}$ is the restriction to $\mathcal{N}$ of some continuous $\pi : (\mathbb{N} \rightharpoonup \mathbb{N}) \to (\mathbb{N} \rightharpoonup \mathbb{N})$, then f is continuous.*

The Corollary makes it possible to recognize continuity of specific functions on Baire space instantly, by inspection, simply noticing that every digit $f(x)(i)$ of each value $f(x)$ can be computed using only finitely many values of x. As in Chapter **6**, a passing remark that some function or other is "evidently continuous" accompanied by no proof typically means an appeal to this result.

10.18. Definition. A pointset $K \subseteq \mathcal{N}$ is **compact** if $K = [T]$ with a tree T on $\mathbb{N}$ which is finitely branching. In particular, every compact pointset is closed, and $\mathcal{C}$ is compact.

Some cheating is involved in adopting this as the definition of compactness for pointsets, since there is a perfectly general definition of compactness for sets in arbitrary topological spaces, by which **10.18** is a theorem. Without comment, we did the same for perfection, which is also a general, topological notion. What we need here are the combinatorial properties of these pointsets specific to Baire space and we have relegated their topological characterizations to the problems, **x10.17** and **x10.21**.

10.19. Proposition. (1) *The image $f[K]$ of a compact pointset K by a continuous function $f : \mathcal{N} \to \mathcal{N}$ is compact.*

(2) *The image $f[K]$ of a compact and perfect pointset K by a continuous injection $f : \mathcal{N} \rightarrowtail \mathcal{N}$ is compact and perfect.*

PROOF. (1) Suppose $K = [T]$ where T is finitely branching and τ computes f as in (10-15), and let

$$S = T^{f[K]} = \{v \mid (\exists x \in K)[v \sqsubseteq f(x)]\}$$

be the tree of initial segments of the image $f[K]$. It is enough to prove that S is a finitely branching tree and $f[K] = [S]$.

To see first that S is finitely branching, suppose $v \in S$, let

$$B =_{\mathrm{df}} \{u \in T \mid v \sqsubseteq \tau(u) \vee v \mid \tau(u)\},$$

and suppose that $x \in [T]$. If $v \mid f(x)$, then for some n, $v \mid \tau(\overline{x}(n))$, so $\overline{x}(n) \in B$; and if $v \sqsubseteq f(x)$, then for some n, $v \sqsubseteq \tau(\overline{x}(n))$, and again $\overline{x}(n) \in B$. Thus, B is a bar for T, and by the Fan Theorem **9.9** it must have a finite subset

$$B_0 = \{u_0, \dots, u_n\} \subseteq B$$

which is also a bar. Thus, for every $x \in K$ such that $v \sqsubseteq f(x)$, there exists some u_i such that $v \sqsubseteq \tau(u_i) \sqsubseteq f(x)$, so that every child of v in S is an initial segment of some $\tau(u_i)$, and there are only finitely many of those.

Clearly, $f[K] \subseteq [S]$. To prove $[S] \subseteq f[K]$, suppose towards a contradiction that $y \in [S] \setminus f[K]$ and let

$$B =_{\mathrm{df}} \{u \in T \mid \tau(u) \mid y\}.$$

Now B is a bar for T, because the only way that $\tau(\overline{x}(n))$ can be compatible with y for every n is if $f(x) = y$. By the Fan Theorem again, there is a finite subset

$$B_0 = \{u_0, \dots, u_n\} \subseteq B$$

which is also a bar for T. Let

$$k = \max\{\mathrm{lh}(\tau(u_i)) \mid i \leq n\} + 1,$$

and choose some $x \in [T]$ such that $\overline{y}(k) \subseteq f(x)$, which exists because $y \in [S]$, so that y can be approximated arbitrarily well by points in the image of f. On the other hand, $u_i \sqsubseteq x$ for some i since B_0 is a bar; hence $\tau(u_i) \sqsubseteq f(x)$ because τ computes f; so both $\tau(u_i)$ and $\overline{y}(k)$ are initial segments of $f(x)$, and hence compatible; and since $\tau(u_i)$ has smaller length than $\overline{y}(k)$, this means that $\tau(u_i) \sqsubseteq y$, which contradicts the definition of B.

(2) With the same notation as in (1) and the additional hypothesis, let $v \in S$, so that for some $u \in T$, $v \sqsubseteq \tau(u)$. Since T is splitting, there exist distinct points

$$x_1, x_2 \in K \cap \mathcal{N}_u,$$

and since τ computes f,

$$\tau(u) \sqsubseteq f(x_1), \quad \tau(u) \sqsubseteq f(x_2). \tag{10-17}$$

But $f(x_1) \neq f(x_2)$, because f is an injection, so there exist incompatible $v_1 \sqsubseteq f(x_1)$, $v_2 \sqsubseteq f(x_2)$, which extend $\tau(u)$ by (10-17), so they split $\tau(u)$ and hence the smaller $v \sqsubseteq \tau(u)$ in S. ⊣

10.20. Perfect Set Theorem (Suslin, 1916). *Every uncountable, analytic set has a non-empty, perfect subset.*

PROOF. Assume that $A = f[\mathbb{N}^*]$ is uncountable, suppose τ computes f, and let

$$T =_{\text{df}} \{u \in \mathbb{N}^* \mid |f[\mathcal{N}_u]| >_c \aleph_0\}. \tag{10-18}$$

Clearly, T is a non-empty tree.

Lemma. *The tree T is τ-splitting, i.e., for each $u \in T$, there exist $u_1, u_2 \in T$ such that*

$$u \sqsubseteq u_1, \quad u \sqsubseteq u_2, \quad \tau(u_1) \mid \tau(u_2).$$

Proof. For any $u \in T$ and any fixed $x \in \mathcal{N}_u$,

$$f[\mathcal{N}_u] = \{f(x)\} \cup \bigcup \{f[\mathcal{N}_{u'}] \mid \tau(u') \mid f(x)\} \tag{10-19}$$

since $f(y) \neq f(x) \Longrightarrow \tau(u') \sqsubseteq f(y)$ for some u' such that $\tau(u')$ is incompatible with $f(x)$. If the Lemma fails at u, then

$$u \sqsubseteq u' \in T \Longrightarrow \tau(u') \sqsubseteq f(x);$$

thus each image $f[\mathcal{N}_{u'}]$ with $\tau(u') \mid f(x)$ in (10-19) involves some $u' \notin T$ and is countable, and there are only countably many choices for u'. Thus, $f[\mathcal{N}_u]$ is the union of a singleton and a countable family of countable sets, hence countable, contrary to hypothesis. ⊣ (Lemma)

As in **10.10**, we choose functions

$$l : T \to T, \quad r : T \to T$$

which witness the τ-splitting property for T, i.e., for each $u \in T$,

$$u \sqsubseteq l(u), \quad u \sqsubseteq r(u), \quad \tau(l(u)) \mid \tau(r(u)),$$

and we define by the String Recursion Theorem **5.33** a function

$$\sigma : \{0, 1\}^* \rightharpoonup T$$

from the tree of all binary strings into T which satisfies the identities

$$\sigma(\emptyset) = \emptyset, \quad \sigma(u \star \langle 0 \rangle) = l(\sigma(u)), \quad \sigma(u \star \langle 1 \rangle) = r(\sigma(u)),$$

and which, as a consequence, is monotone. The key property of this σ is that it takes incompatible binary strings to τ-incompatible strings,

$$u \mid v \Longrightarrow \tau(\sigma(u)) \mid \tau(\sigma(v)), \tag{10-20}$$

and it is verified exactly as (10-11) was verified in the proof of **10.10**. In addition, σ computes a continuous $g : \mathcal{C} \to \mathcal{N}$,

$$g(x) = \sup\{\sigma(u) \mid u \sqsubseteq x\},$$

and evidently

$$g[\mathcal{C}] \subseteq [T]. \tag{10-21}$$

Consider now the composition $h = fg$ of the given f and this g, which is computed by the composition of τ and σ:

$$h(x) = \sup\{\tau(\sigma(u)) \mid u \sqsubseteq x\}.$$

This is continuous and injective by (10-20), so its image $h[\mathcal{C}] = fg[\mathcal{C}]$ is compact and perfect by **10.19**, and it is included in $f[T] \subseteq A$ by (10-21). $\dashv$

The result means nothing, of course, until we prove that there are lots and lots of analytic sets.

10.21. Lemma. *Every closed pointset is analytic.*

PROOF. Let $T = T^F$ as in (10-8) for the given closed set $F \neq \emptyset$, so in addition to $F = [T]$ we also know that every string in T has an extension, i.e., there are no terminal nodes. Thus, we can fix a function $l : T \to T$ such that

$$u \in T \Longrightarrow u \sqsubseteq l(u) \;\&\; \mathrm{lh}(l(u)) = \mathrm{lh}(u) + 1.$$

Let also

$$\mathrm{rtail}(u) =_{\mathrm{df}} u \restriction [0, \mathrm{lh}(u) - 1) \quad (\mathrm{lh}(u) > 0) \tag{10-22}$$

be the partial function which strips each non-empty string of its last element. By the String Recursion Theorem **5.33**, there is a function $\tau : \mathbb{N}^* \rightharpoonup T$ such that

$$\tau(u) = \begin{cases} u, & \text{if } u \in T, \\ l(\tau(\mathrm{rtail}(u))), & \text{if } u \notin T, \end{cases}$$

which is (easily) a *projection* of $\mathbb{N}^*$ onto T, i.e., it is total, length preserving and the identity on T, and which (as a consequence) computes a function $f : \mathcal{N} \twoheadrightarrow [T]$. $\dashv$

10.22. Lemma. *Every continuous image of an analytic pointset is analytic.*

PROOF. If $A = f[B]$ and $B = g[\mathcal{N}]$, then $A = fg[\mathcal{N}]$, and the composition fg is continuous. $\dashv$

10.23. Lemma. *If $f, g : \mathcal{N} \to \mathcal{N}$ are continuous functions, then the set*

$$E = \{x \mid f(x) = g(x)\}$$

of points on which they agree is closed.

PROOF. Because distinct points can be approximated by incompatible initial segments,

$$\begin{aligned} x \notin E &\iff f(x) \neq g(x) \\ &\iff (\exists u, v)[f(x) \in \mathcal{N}_u \;\&\; g(x) \in \mathcal{N}_v \;\&\; u \mid v], \end{aligned}$$

which means that

$$cE = \bigcup \{f^{-1}[\mathcal{N}_u] \cap g^{-1}[\mathcal{N}_v] \mid u \mid v\},$$

so that cE is the union of open sets and hence open. ⊣

10.24. Theorem. *Countable unions and countable intersections of analytic pointsets are analytic.*

PROOF. Suppose that $A_n = f_n[\mathcal{N}]$ with each f_n continuous and define first $f : \mathcal{N} \to \mathcal{N}$ by the formula

$$f(z) = f_{z(0)}(\mathrm{tail}(z)),$$

where

$$\mathrm{tail}(z) =_{\mathrm{df}} (i \mapsto z(i+1)) = (z(1), z(2), \dots)$$

is the function which decapitates points. Evidently f is continuous: because each $f(z)(i)$ can be computed from finitely many values of z, first setting $n = z(0)$ and then using the finitely many values of $\mathrm{tail}(z)$ needed to compute $f_n(\mathrm{tail}(z))$. Moreover:

$$\begin{aligned} y \in \textstyle\bigcup_n f_n[\mathcal{N}] &\iff (\exists n \in \mathbb{N}, x \in \mathcal{N})[y = f_n(x)] \\ &\iff (\exists z \in \mathcal{N})[y = f_{z(0)}(\mathrm{tail}(z))] \\ &\qquad\qquad\qquad \text{taking } z(0) = n, \mathrm{tail}(z) = x \\ &\iff (\exists z \in \mathcal{N})[y = f(z)], \end{aligned}$$

so $\bigcup_n A_n = f[\mathcal{N}]$ and the union of the A_n's is analytic.

The key fact for this argument was that the mapping

$$z \mapsto (z(0), \mathrm{tail}(z))$$

is a surjection of $\mathcal{N}$ onto $\mathbb{N} \times \mathcal{N}$—actually a bijection—with continuous components. To prove that the intersection $\bigcap_n A_n$ is analytic, we need a similar surjection

$$\pi : \mathcal{N} \twoheadrightarrow (\mathbb{N} \to \mathcal{N})$$

of $\mathcal{N}$ onto the set of infinite sequences of points. To construct such a π, fix some bijection $\rho : \mathbb{N} \times \mathbb{N} \rightarrowtail\!\!\!\!\!\twoheadrightarrow \mathbb{N}$ and set

$$\rho_n(z) = (i \mapsto z(\rho(n, i))). \tag{10-23}$$

Each $\rho_n : \mathcal{N} \to \mathcal{N}$ is clearly continuous and for each infinite sequence $\{x_n\}_{n \in \mathbb{N}}$ of points we can define z such that

$$z(\rho(n, i)) = x_n(i) \quad (n, i \in \mathbb{N}),$$

so that

$$\rho_n(z) = x_n \quad (n \in \mathbb{N});$$

in other words, the mapping

$$\pi(z) = (n \mapsto \rho_n(z))$$

is a surjection. Using $\mathbf{AC}_{\mathbb{N}}$ now,

$$\begin{aligned} y \in \bigcap_n A_n &\iff (\forall n)(\exists x)[y = f_n(x)] \\ &\iff (\exists \{x_n\}_{n \in \mathbb{N}})(\forall n)[y = f_n(x_n)] \\ &\iff (\exists z \in \mathcal{N})(\forall n)[y = f_n(\rho_n(z))] \\ &\iff (\exists z \in \mathcal{N})[(\forall n)[f_n(\rho_n(z)) = f_0(\rho_0(z))] \\ &\qquad\qquad \& \; y = f_0(\rho_0(z))]. \end{aligned} \tag{10-24}$$

For each n, the set

$$B_n = \{z \in \mathcal{N} \mid f_n(\rho_n(z)) = f_0(\rho_0(z))\}$$

is closed by **10.23**, hence the intersection

$$B = \bigcap_n B_n$$

is also closed. From (10-24), however,

$$\bigcap_n A_n = f_0 \rho_0[B],$$

which means that the intersection of the A_n's is analytic. ⊣

10.25. Definition. The family $\mathcal{B}(X)$ of the **Borel** subsets of a topological space X is the smallest family of subsets of X which includes the open sets and is a **σ-field**, i.e., it is closed under countable unions and complementation:

$$\begin{aligned} (\forall n)[A_n \in \mathcal{B}(X)] &\implies \bigcup_n A_n \in \mathcal{B}(X), \\ A \in \mathcal{B}(X) &\implies cA \in \mathcal{B}(X). \end{aligned}$$

We are mostly interested in Baire space of course,

$$\mathcal{B} =_{\mathrm{df}} \mathcal{B}(\mathcal{N}) = \text{the family of Borel pointsets.}$$

10.26. Exercise. *Prove that the definition makes sense, i.e., the intersection*

$$\begin{aligned} \mathcal{B}(X) = \bigcap \{\mathscr{E} \mid \mathcal{G} \subseteq \mathscr{E} \\ & \; (\forall \{A_n\}_n \subseteq \mathscr{E})[\textstyle\bigcup_n A_n \in \mathscr{E}] \\ & \; (\forall A \in \mathscr{E})[cA \in \mathscr{E}]\} \end{aligned}$$

is a σ-field which contains the open sets, and hence the least such.

10.27. Exercise. *The intersection $\bigcap_n A_n$ of every sequence of Borel sets $\{A_n\}$ is a Borel set.*

10.28. Corollary. *Every Borel pointset is analytic* (Suslin) *and hence has property* **P** (Alexandroff, Hausdorff).

PROOF. Let

$$C\mathcal{A} = \{A \subseteq \mathcal{N} \mid cA \in \mathcal{A}\} \tag{10-25}$$

be the family of **co-analytic pointsets**, those with analytic complements. The family $\mathcal{A} \cap C\mathcal{A}$ of pointsets which are both analytic and co-analytic is a

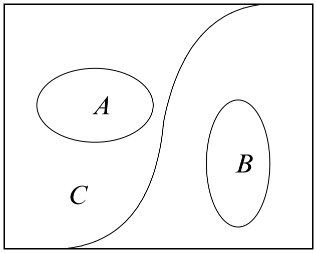

FIGURE **10.2**. C separates A from B.

σ-field, since it is closed under complementation by definition, and if each $A_n \in \mathcal{A} \cap C\mathcal{A}$, then $\bigcup_n A_n$ and

$$c(\bigcup_n A_n) = \bigcap_n cA_n$$

are both analytic by the theorem. Since every open set

$$G = \bigcup_n \{\mathcal{N}_u \mid \mathcal{N}_u \subseteq G\}$$

is a countable union of neighborhoods, hence analytic, and also co-analytic by **10.21**, $\mathcal{A} \cap C\mathcal{A}$ is a σ-field which contains all the open pointsets and hence includes every Borel set. $\dashv$

In the next two theorems we clarify somewhat the relation between analytic and Borel pointsets.

A pointset $C \subseteq \mathcal{N}$ *separates* a pointset A from another pointset B, if

$$A \subseteq C, \quad C \cap B = \emptyset.$$

Notice that if some C separates A from B, then $A \cap B = \emptyset$.

10.29. Lemma. (1) *If $\{A_i\}$, $\{B_j\}$ are two sequences of pointsets and for all i, j, C_{ij} separates A_i from B_j, then the set $C = \bigcup_i \bigcap_j C_{ij}$ separates $\bigcup_i A_i$ from $\bigcup_j B_j$, i.e.,*

$$\bigcup_i A_i \subseteq \bigcup_i \bigcap_j C_{ij}, \quad \Big(\bigcup_i \bigcap_j C_{ij}\Big) \cap \bigcup_j B_j = \emptyset. \tag{10-26}$$

(2) *If $\{A_i\}$, $\{B_j\}$ are two sequences of pointsets and no Borel set separates $A = \bigcup_i A_i$ from $B = \bigcup_j B_j$, then there exist two numbers i_0 and j_0 such that no Borel set separates A_{i_0} from B_{j_0}.*

PROOF. (1) For any fixed i and all j, by hypothesis, $A_i \subseteq C_{ij}$, and so

$$A_i \subseteq \bigcap_j C_{ij};$$

so taking the unions of both sides, we get

$$A = \bigcup_i A_i \subseteq \bigcup_i \bigcap_j C_{ij},$$

which is the first inclusion claimed by the Lemma. For the second, we notice that the hypothesis $B_j \cap C_{ij} = \emptyset$ means exactly that

$$B_j \subseteq cC_{ij} \quad (cC_{ij} = \mathcal{N} \setminus C_{ij});$$

and so, fixing i and taking unions again, we get

$$B = \bigcup_j B_j \subseteq \bigcup_j C_{ij},$$

which, since i was arbitrary, yields

$$B \subseteq \bigcap_i \bigcup_j c C_{ij}.$$

Now Problem **x1.3** (De Morgan's laws) yields

$$\bigcap_i \bigcup_j c C_{ij} = c\Big(\bigcup_i \bigcap_j C_{ij}\Big),$$

so that $B \cap \Big(\bigcup_i \bigcap_j C_{ij}\Big) = \emptyset$, which was what we needed to prove.

(2) follows easily, by contradiction: because if some Borel set C_{ij} separated each A_i from each B_j, then $\bigcup_i \bigcap_j C_{ij}$ separates A from B—and it is a Borel set. ⊣

10.30. The Separation Theorem (Lusin) *If $A, B \subseteq \mathcal{N}$ are analytic pointsets and $A \cap B = \emptyset$, then there exists a Borel pointset C which separates A from B.*

PROOF. Suppose $A = f[\mathcal{N}], B = g[\mathcal{N}]$, where f, g are continuous functions which by Theorem **10.15** are computed by given, monotone string functions $\sigma, \tau : \mathbb{N}^* \to \mathbb{N}^*$,

$$f(x) = \lim_n \sigma(\overline{x}(n)), \quad g(y) = \lim_n \tau(\overline{y}(n)).$$

For any two strings u, v, put

$$A_u = f[\mathcal{N}_u] = \{f(x) \mid u \sqsubseteq x\}, \quad B_v = g[\mathcal{N}_v] = \{g(y) \mid v \sqsubseteq y\},$$

and record the fact that

$$A_u \subseteq \mathcal{N}_{\sigma(u)}, \quad B_v \subseteq \mathcal{N}_{\tau(v)}, \tag{10-27}$$

which follows directly from what it means for σ to compute f and τ to compute g. Notice also that $A_\emptyset = A, B_\emptyset = B$, and, easily, for all u, v,

$$A_u = \bigcup_i A_{u \star \langle i \rangle}, \quad B_v = \bigcup_j B_{v \star \langle j \rangle}. \tag{10-28}$$

We now assume towards a contradiction that no Borel set separates A from B and we apply repeatedly Lemma **10.29**, using (10-28): since no Borel set separates $A = A_\emptyset$ from $B = B_\emptyset$, there exist numbers i_0, j_0 such that no Borel set separates $A_{\langle i_0 \rangle}$ from $B_{\langle j_0 \rangle}$; and so, there exist i_1, j_1 such that no Borel set separates $A_{\langle i_0, i_1 \rangle}$ from $B_{\langle j_0, j_1 \rangle}$; and so, etc. Formally, we define recursively two sequences of numbers

$$x = (i_0, i_1, \dots), \; y = (j_0, j_1, \dots) \in \mathcal{N}$$

such that for all n, *no Borel set separates $A_{\overline{x}(n)}$ from $B_{\overline{y}(n)}$*. Now this means that,

$$\text{for all } n, \mathcal{N}_{\sigma(\overline{x}(n))} \cap \mathcal{N}_{\tau(\overline{y}(n))} \neq \emptyset, \tag{10-29}$$

since otherwise $\mathcal{N}_{\sigma(\overline{x}(n))}$ would separate $A_{\overline{x}(n)}$ from $B_{\overline{y}(n)}$, by (10-27); and, finally, (10-29) says directly that

$$f(x) = g(y) \in \bigcap_n \Big(\mathcal{N}_{\sigma(\overline{x}(n))} \cap \mathcal{N}_{\tau(\overline{y}(n))}\Big),$$

which contradicts the hypothesis, that $A \cap B = \emptyset$. ⊣

10.31. Suslin's Theorem. *A pointset $A \subseteq \mathcal{N}$ is Borel if and only if both A and its complement cA are analytic.*

PROOF of the non-trivial direction is immediate, by applying the Separation Theorem to A and cA. ⊣

Suslin introduced the family of analytic pointsets in 1917 and proved a slew of theorems about it, including the Perfect Set Theorem **10.20**, his famous characterization **10.31**, and that *not every analytic pointset is Borel*, which we will not prove here.[26] The Borel sets had been introduced more than a decade earlier by Borel and Lebesgue and they were the key to the successful development of the theory of *Lebesgue integration*, one of the chief achievements of 19th century analysis. For most purposes of integration theory, including its later, fundamental applications to *probability*, every pointset of interest is *almost equal* to a Borel set, in a precise sense which basically allows us to study the subject as if every pointset were Borel. Because of this, the Continuum Problem for Borel sets was considered very important, and its simultaneous, independent solutions published by Alexandroff and Haussdorff in 1916 (just before Suslin established the more general Theorem **10.20**) were celebrated as a major achievement.

The family of analytic sets falls far short from exhausting the powerset of $\mathcal{N}$, Problem **x10.9**. Still, one might hope that the method we used to solve the Continuum Problem for them might be extended to prove the full Continuum Hypothesis, but this too is far from the mark.

10.32. Theorem. (**AC**) *There exists a pointset $A \subset \mathcal{N}$ which is uncountable but contains no non-empty perfect set.*

PROOF. The key fact is that there are exactly as many non-empty, perfect sets as there are points in $\mathcal{N}$:

Lemma 1. *If* $\mathscr{P} = \{P \subseteq \mathcal{N} \mid P \neq \emptyset, P \text{ perfect}\}$, *then* $|\mathscr{P}| =_c \mathfrak{c}$.

Proof. For each $y \in \mathcal{N}$, the pointset

$$A_y = \{x \mid (\forall n)[y(n) \leq x(n)\}$$

is easily perfect, equally easily $y \neq z \Longrightarrow A_y \neq A_z$, hence $\mathfrak{c} =_c |\mathcal{N}| \leq_c |\mathscr{P}|$. On the other hand, each perfect set $P = [T^P]$ is the body of a tree on $\mathbb{N}$ which determines it, so

$$|\mathscr{P}| \leq_c |\mathcal{P}(\mathbb{N}^*)| =_c |\mathcal{P}(\mathbb{N})| =_c \mathfrak{c}. \qquad \dashv \text{ (Lemma 1)}$$

Fix a set

$$I =_c \mathfrak{c} =_c \mathscr{P}, \tag{10-30}$$

[26] The study of analytic and Borel pointsets is the core of *Descriptive Set Theory*, one of the most beautiful parts of our subject, but (unfortunately) beyond the scope of these Notes.

for example $I = \mathfrak{c}$, and bijections

$$\alpha \mapsto x_\alpha \in \mathcal{N}, \quad \alpha \mapsto P_\alpha \in \mathscr{P} \quad (\alpha \in I)$$

which witness the equinumerosities (10-30). Fix also a best wellordering $\leq$ of I. We will define by transfinite recursion on $(I, \leq)$ injections

$$f_\alpha : \mathbf{seg}(\alpha) \rightarrowtail\!\!\!\!\!\twoheadrightarrow A_\alpha \subset \mathcal{N},\ g_\alpha : \mathbf{seg}(\alpha) \rightarrowtail\!\!\!\!\!\twoheadrightarrow B_\alpha \subset \mathcal{N} \quad (\alpha \in I),$$

so that the following conditions hold.

(1) If $\alpha \leq \beta$, then $f_\alpha \subseteq f_\beta, g_\alpha \subseteq g_\beta$, so that $A_\alpha \subseteq A_\beta$ and $B_\alpha \subseteq B_\beta$.
(2) For each $\alpha \in I$, $A_\alpha \cap B_\alpha = \emptyset$.
(3) For each $\alpha \in I$, $B_{S\alpha} \cap P_\alpha \neq \emptyset$, where S is the successor function in the well ordered set $(I, \leq)$.

Lemma 2. *If* f_α, g_α $(\alpha \in I)$ *satisfy* (1) – (3), *then*

$$A = \bigcup_{\alpha \in I} A_\alpha =_c \mathfrak{c},$$

but A *has no non-empty, perfect subset.*

Proof. That $A =_c I =_c \mathfrak{c}$ follows immediately from (1), since

$$\bigcup_\alpha f_\alpha : I \rightarrowtail\!\!\!\!\!\twoheadrightarrow A \text{ and } I =_c \mathfrak{c}.$$

For the second claim, the key observation is that

$$A \cap B_\beta = \emptyset \quad (\beta \in I);$$

this is because if $x \in A_\alpha \cap B_\beta$, then with $\gamma = \max\{\alpha, \beta\}$, by (1), $x \in B_\gamma \cap B_\gamma$, contradicting (2). Now if $P \neq \emptyset$ is perfect, then $P = P_\alpha$ for some $\alpha \in I$, hence there exists some $x \in P_\alpha \cap B_{S\alpha}$ and then $x \notin A$, so $P_\alpha \nsubseteq A$. ⊣ (Lemma 2)

The definitions of f_α, g_α are practically forced on us by conditions (1) – (3). We outline the proofs of (1) – (3) by transfinite induction together with the clauses of the transfinite recursion definition; pedantically the proof should be separated out and explained after the definition is completed.

(a) At the minimum 0 of I, set $f_0 = g_0 = \emptyset$.

(b) If λ is a limit point of I, set

$$f_\lambda = \bigcup_{\alpha<\lambda} f_\alpha, \quad g_\lambda = \bigcup_{\alpha<\lambda} g_\alpha.$$

Conditions (1) and (2) follow immediately from the induction hypothesis, and (3) is not relevant to this case.

(c) Suppose $\beta = S\alpha$ is a successor point in I. By the induction hypothesis, A_α and B_α are equinumerous with $\mathbf{seg}(\alpha)$ and $\mathbf{seg}(\alpha) <_c I$, because $\leq$ is a best wellordering. Thus, $|A_\alpha|$, $|B_\alpha|$ are both smaller than $\mathfrak{c}$, hence $|A_\alpha \cup B_\alpha| <_c \mathfrak{c}$, and we can find in the non-empty, perfect set $P_\alpha =_c \mathcal{N}$ distinct points

$$x, y \in P_\alpha \setminus (A_\alpha \cup B_\alpha);$$

we set

$$f_\beta = f_\alpha \cup \{(\alpha, x)\}, \quad g_\beta = g_\alpha \cup \{(\alpha, y)\},$$

and (1) – (3) follow easily. ⊣

The construction obviously proves more than what is claimed in the theorem: $|A| =_c \mathfrak{c}$ and both A and its complement cA intersect every non-empty, perfect set. We leave for the problems some additional variations which make it even more obvious that the program of proving the Continuum Hypothesis by using the Cantor-Bendixson Theorem is hopeless.

Actually, it is not only this program for settling the Continuum Problem which fails: every attempt to prove or disprove **CH** from the axioms of **ZDC**+**AC** is doomed, by the following two central independence results.

10.33. Consistency of GCH, the Generalized Continuum Hypothesis (Gödel, 1939). The model L of constructible sets satisfies the Generalized Continuum Hypothesis **GCH**, (9-7), so, in particular, the Continuum Hypothesis cannot be refuted in **ZDC**+**AC**.

10.34. Independence of the Continuum Hypothesis CH (Cohen, 1963). There is a model of **ZDC**+**AC** in which the Continuum Hypothesis fails, hence **CH** cannot be proved in **ZDC**+**AC**. Cohen's forcing model can be modified in many ways to manipulate the cardinalities of pointsets and subsets of larger powersets.

10.35. What does the independence of CH mean? Both the Gödel and the Cohen methods of proof are very robust and they have been adapted to show that the Continuum Problem cannot be settled on the basis of many reasonable and plausible strengthenings of **ZDC**+**AC** by additional axioms. The same is true of the Axiom of Choice, of course, or the Axiom of Infinity for that matter, but it is clear that these propositions express new, fundamental set theoretic principles which are most likely true but cannot (and, in fact, cannot be expected to) be proved from simpler axioms by logic alone. The Continuum Hypothesis has the look of a technical, mathematical problem which should be settled definitively by a proof, but we seem to lack the insight needed to divine the necessary axioms.

Much has been made of this independence of **CH** (and many more assertions about sets) from variants of the known axioms of set theory, and some have used it to argue against any objective reality behind the "formal", axiomatic results of the subject. Using the method of *arithmetization* introduced by Gödel, however, we can translate questions about the *existence of proofs* into precise, technical conjectures about integers: since there exist such conjectures[27] which (like **CH**) can be shown to be undecidable in the known, plausible axiomatic theories, are we then forced to deny objective reality to

[27] The type of statements we have in mind here are of the form "*if* **ZDC**+**AC** *is consistent, then so is T*", where T is some strong extension of **ZDC**+**AC** which, in fact, implies the consistency of **ZDC**+**AC**. Gödel's *Second Incompleteness Theorem* implies that statements of this type are independent of **ZDC**+**AC** (unless **ZDC**+**AC** is inconsistent), and there are many of them about whose truth there is genuine controversy.

the natural numbers? In fact, it is not possible to discuss such problems intelligently without reference to notions and results of Mathematical Logic which are beyond the scope of these Notes, and we will resist the temptation.

Incidentally, there are scores of interesting propositions about sets which cannot be settled on the basis of **ZDC** or **ZDC+AC**; **CH** is only the most interesting of them. We mention here just three more independence results of this type, because they are relevant to the Perfect Set Theorem **10.20**.

10.36. (Gödel, 1939) In the model L of constructible sets, there exists an uncountable, co-analytic set which has no proper perfect subset. This means that we cannot improve the Perfect Set Theorem **10.20** in **ZDC+AC** to show that every co-analytic pointset has property **P**.

10.37. (Solovay, 1970) There is a model of **ZDC+AC** in which every "definable" pointset has property **P**. We will not attempt to define "definable", but analytic complements are definable.

10.38. (Solovay, 1970) There is a model of **ZDC** in which every pointset has property **P**.

Solovay's models are constructed by Cohen's forcing method, but like Gödel's L, they have many more canonical properties which yield numerous unprovability results. The first Solovay model witnesses (with **10.36**) that the property **P** for analytic complements cannot be proved or refuted in **ZDC+AC**. The second Solovay model shows that **ZDC** cannot prove the existence of an uncountable pointset with no non-empty, perfect subset; **DC** is not a sufficiently strong choice principle to effect the construction.

Problems for Chapter 10

x10.1. Prove that if $F \subseteq \mathcal{N}$ is closed, then there is a unique tree T on $\mathbb{N}$ without terminal nodes such that $F = [T]$, namely the tree T^F defined in (10-8).

x10.2. Prove that the decomposition (10-12) of a closed pointset F into a perfect set P and a countable set S determines uniquely P and S.

x10.3. Give an example of a closed pointset $F \subseteq \mathcal{N}$ and a continuous $f : \mathcal{N} \to \mathcal{N}$, such that the image $f[F]$ is not closed.

x10.4. Prove that every open pointset is an $\mathcal{F}_\sigma$ and every closed pointset is a $\mathcal{G}_\delta$. The definitions are reviewed in Footnote 25.

***x10.5.** Prove that the inverse image $g^{-1}[A]$ of an analytic pointset A by a continuous function $g : \mathcal{N} \to \mathcal{N}$ is analytic. HINT. Aim for an equivalence of the form

$$y \in g^{-1}[A] \iff (\exists x)[y = f(\rho_1(x)) = g(\rho_2(x))]$$

where f is continuous and ρ_n are defined by (10-23), and then use **10.23**.

*__x10.6.__ Prove that

$$\mathcal{N} = \bigcup_{i \in \mathbb{N}} A_i \Longrightarrow (\exists i)[A_i =_c \mathcal{N}],$$

i.e., $\mathcal{N}$ is not the union of a countable sequence of pointsets of smaller cardinality. HINT. This follows easily from König's Theorem **9.21**, but the relevant special case does not need the full Axiom of Choice.

x10.7. (**AC**) Prove that for every $\kappa \leq_c \mathfrak{c}$, there exists a pointset A with $|A| =_c \kappa$ which contains no non-empty, perfect subset.

x10.8. (**AC**) Prove that there exists an uncountable pointset A such that neither A nor its complement contain an uncountable Borel set.

x10.9. Prove that there are $\mathfrak{c}$-many analytic and Borel pointsets,

$$|\mathcal{A}| =_c |\mathcal{B}| =_c \mathfrak{c}.$$

10.39. Definition. A function $f : X \to Y$ from one topological space into another is **Borel measurable** if the inverse image $f^{-1}[G]$ of every open subset of Y is a Borel subset of X.

x10.10. The composition $gf : X \to Z$ of two Borel measurable functions $f : X \to Y$ and $g : Y \to Z$ is Borel measurable.

x10.11. The inverse image $f^{-1}[A]$ of a Borel set $A \subseteq Y$ by a Borel measurable function $f : X \to Y$ is a Borel subset of X.

10.40. Definition. Two topological spaces X, Y are **Borel isomorphic** if there exists a Borel measurable bijection $f : X \rightarrowtail\!\!\!\!\twoheadrightarrow Y$, whose inverse $f^{-1} : Y \rightarrowtail\!\!\!\!\twoheadrightarrow X$ is also Borel measurable. Borel isomorphic spaces have the same measure-theoretic structure and for all practical purposes can be "identified" in measure theory.

*__x10.12.__ Suppose $f : X \rightarrowtail Y$ and $g : Y \rightarrowtail X$ are Borel measurable injections between topological spaces, with the following additional property:[28] there exists Borel measurable functions $f_1 : Y \to X$ and $g_1 : X \to Y$ which are inverses of f and g in the sense that

$$f_1 f(x) = x \quad (x \in X),$$
$$g_1 g(y) = y \quad (y \in Y).$$

Prove that X and Y are Borel isomorphic. HINT. Use the proof of the Schröder-Bernstein Theorem **2.26**.

x10.13. Consider the Cantor set $\mathcal{C}$ as a topological subspace of $\mathcal{N}$ in the obvious way, the open sets being unions of neighborhoods of the form

$$\mathcal{N}_u = \{x \in \mathcal{C} \mid u \sqsubseteq x\} \quad (u \in \{0,1\}^*).$$

Prove that $\mathcal{C}$ and $\mathcal{N}$ are Borel isomorphic.

[28] In fact, every Borel injection has this property, but the proof of this requires some work.

In the remaining problems we explore the connection of the specific, combinatorial notions we studied in Baire space with their general, topological versions.

10.41. Definition. A point x is a **limit point** of a set A in a topological space X if every open set G which contains x contains also some point of A other than x,

$$(\forall G)[G \text{ open and } x \in G \Longrightarrow (\exists y \in A \cap G)[x \neq y]].$$

A limit point of A may or may not be a member of A. *A point of A which is not a limit point of A is* **isolated in** A.

x10.14. Determine the limit points and the isolated points of the pointset

$$B = \{x \in \mathcal{N} \mid x(0) = 1 \vee (\forall n)[x(n) = 2] \vee (\exists n)[x(n) = 3]\}.$$

x10.15. Prove that x is a limit point of A if and only if every open set containing x contains infinitely many points of A.

x10.16. Prove that a set is closed in a topological space X if and only if it contains all its limit points.

x10.17. Prove that a pointset P is perfect if and only if it is closed and has no isolated points, i.e., every point of P is a limit point of P. This equivalence identifies the specific definition of perfect pointsets we adopted with the classical, topological definition.

10.42. Definition. A sequence $(n \mapsto x_n)$ of points in a topological space X **converges** to a point x or has x as its **limit** if every open set containing x contains all but finitely many of the terms of the sequence,

$$\lim_n x_n = x \iff_{\text{df}} (\forall G \text{ open}, x \in G)(\exists n \in \mathbb{N})(\forall i \geq n)[x_i \in G].$$

x10.18. Prove that a point x is a limit point of a pointset A if and only if $x = \lim_n x_n$ is the limit of some sequence $(n \mapsto x_n \in A)$ of points in A. Which choice principle did you use, if any?

x10.19. Prove that a function $f : \mathcal{N} \to \mathcal{N}$ is continuous if and only if

$$f(\lim_n x_n) = \lim_n f(x_n),$$

whenever $\lim_n x_n$ exists. Which choice principle did you use, if any?

x10.20. *A topological space X is* **Hausdorff** *if for any two points $x \neq y$, there exist disjoint open sets $G \cap H = \emptyset$ such that $x \in G$ and $y \in H$.* Prove that if $f, g : X \to Y$ are continuous functions and Y is Hausdorff, then the set $\{x \in X \mid f(x) = g(x)\}$ is closed in X.

10.43. Definition. An **open covering** of a set K in a topological space X is any family $\mathscr{G}$ of open sets whose union includes K, $K \subseteq \bigcup \mathscr{G}$. A set K is **compact** in X if every open covering of K includes a finite subcovering, i.e., for every family $\mathscr{G}$ of open sets,

$$K \subseteq \bigcup \mathscr{G} \Longrightarrow (\exists G_0, \dots, G_n \in \mathscr{G})[K \subseteq \bigcup_{n \leq i} G_i].$$

*__x10.21.__ Prove that a pointset is compact by Definition **10.18** if and only if it is compact by Definition **10.43**. HINT. You will need König's Lemma **9.7**.

x10.22. Prove that for any two topological spaces X, Y, any continuous function $f : X \to Y$ and any compact set $K \subseteq X$, the image $f[K]$ is compact in Y.